MATHS MANTHAN
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Subject: Mathematics | **Periodic Test Paper 09 (Cumulative)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

General Instructions:

- This question paper consists of **12 questions** divided into **Five Sections**: A, B, C, D, and E.
- **Section A** comprises **5 questions** of **1 mark** each (Q1–Q4 MCQs, Q5 Assertion-Reason).
- **Section B** comprises **2 questions** (Q6–Q7) of **2 marks** each.
- **Section C** comprises **2 questions** (Q8–Q9) of **3 marks** each.
- **Section D** comprises **2 questions** (Q10–Q11) of **5 marks** each.
- **Section E** comprises **1 Applied Case Study** (Q12) of **5 marks** with internal choice in sub-part (iii).
- All questions are compulsory. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice & Assertion-Reason Questions (5 × 1 = 5 Marks)

Q1. In an AP, if $a_n = 3n + 4$, then the first term a_1 is: [1]

(A) 3 (B) 4 (C) 7 (D) 10

Q2. The common difference of the AP whose n^{th} term is given by $a_n = 5 - 2n$ is: [1]

(A) 5 (B) 2 (C) -2 (D) 3

Q3. The 10th term of the AP $2, 7, 12, \ldots$ is: [1]

(A) 47 (B) 42 (C) 52 (D) 37

Q4. If the nth term of an AP is $7 - 4n$, then its common difference is: [1]

(A) 7 (B) 4 (C) -4 (D) 3

Q5. **Assertion (A):** In an arithmetic progression, $a_{20} - a_{10} = 10d$.
Reason (R): The general term of an AP is given by $a_n = a + (n-1)d$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.
(B) Both (A) and (R) are true, but (R) is not correct explanation.
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. Determine the value of k so that $k^2 + 4k + 8$, $2k^2 + 3k + 6$, and $3k^2 + 4k + 4$ are three consecutive terms of an AP. [2]

Q7. If the 7th term of an AP is $\frac{1}{9}$ and its 9th term is $\frac{1}{7}$, find its 63rd term. [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. Find the middle term of the arithmetic progression $6, 13, 20, \ldots, 216$. [3]

Q9. Justify whether -150 is a term of the AP $11, 8, 5, 2, \ldots$. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. The ratio of the sums of m and n terms of an AP is $m^2 : n^2$. Show that the ratio of its mth and n^{th} terms is:

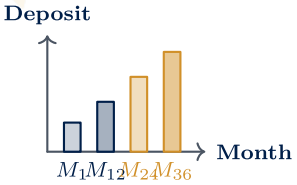
$$\frac{a_m}{a_n} = \frac{2m-1}{2n-1}$$

[5]

Q11. A contract on construction job specifies a penalty for delay of completion beyond a certain date: Rs. 200 for the first day, Rs. 250 for the second day, Rs. 300 for the third day, and so on, the penalty for each succeeding day being Rs. 50 more than for the preceding day. How much money must the contractor pay as penalty, if he has delayed the work by 30 days? [5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: Systematic Monthly Savings Plan (Recurring Deposit)



Aarav begins his financial planning upon turning 18 by opening a monthly recurring deposit savings account at a commercial bank.
He starts with an initial deposit of Rs. 1,500 in the first month. Committed to disciplined investing, he increases his deposit by a uniform amount of Rs. 250 every consecutive month for a tenure of 3 years (36 months).

Based on the above financial scenario, answer the following questions:

(i) Find the amount deposited by Aarav in the 12th month (a_{12}). [1]

(ii) Find the amount deposited by Aarav in the 24th month (a_{24}). [1]

(iii) Calculate the total cumulative savings accumulated by Aarav at the end of 36 months (S_{36}). [3]

OR

(iv) In which month will his monthly deposit first reach Rs. 10,000? [3]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★